

Connectors and Cables -- Guided Notes ANSWER KEY

Unit tp-2.5 | CompTIA Network+ N10-009 Obj. FC0-U71 2.5 | N10-008 FC0-U71 2.5

For instructor use / distribute after students complete the notes.

Question 1

USB Type-A connectors are the _____ (flat rectangular) end that plugs into the _____, such as a computer tower or charger. USB Type-B is a square-ish connector used on _____ such as printers. USB Type-C is reversible and used for both _____ and _____ in modern devices.

Answer: Host; data and power (charging).

Question 2

HDMI cables carry both _____ and _____ signals in a single cable. HDMI 1.4 supports up to 4K at _____ Hz. HDMI 2.0 supports 4K at _____ Hz. DisplayPort can support higher _____ rates than HDMI, making it preferred for high-refresh gaming monitors.

Answer: Video and audio; 30 Hz; 60 Hz; refresh.

Question 3

A Thunderbolt 3 or 4 port uses the same physical _____ connector as USB-C but supports speeds up to _____ Gbps, can carry _____ (video output), _____ (USB protocols), and even PCIe data. Thunderbolt is identified by the _____ icon next to the port.

Answer: USB-C; 40 Gbps; DisplayPort (video); USB; lightning bolt.

Question 4

RJ-45 is the connector used for _____ networking cables (Ethernet). It has _____ pins. RJ-11 is used for _____ lines and has _____ pins. A _____ cable is used to connect a computer directly to another computer without a switch.

Answer: Wired (Ethernet); 8 pins; telephone (DSL/POTS); 6 pins; crossover.

Question 5

3.5mm audio jacks are color coded: _____ (lime green) is for headphones/speakers (output), _____ (pink or red) is for microphone (input), and _____ (blue) is for line-in from an external audio source. The jack carries an _____ (digital/analog) audio signal.

Answer: Green; pink/red; blue; analog.

Question 6

SATA data cables are _____ (wide/narrow) L-shaped connectors carrying only _____ between the motherboard and drive. SATA power cables come from the _____ and carry 3.3V, 5V, and 12V rails. Molex connectors are an older _____ (data/power) connector used for fans and older peripherals.

Answer: Narrow; data; PSU (power supply); power.

Question 7

The DB9 (DE-9) connector is a 9-pin serial port used for _____ devices such as older modems and network equipment console connections. The DB15 (DE-15) is a _____ pin VGA video connector. Both are examples of _____-style connectors, named for their D-shaped metal shell.

Answer: Legacy serial; 15; D-sub.

Question 8 -- Real World Application

REAL WORLD: A student wants to connect a new 4K 144 Hz gaming monitor using a cable they found in a drawer. The cable is HDMI 1.4. They plug it in and the monitor only runs at 4K 30 Hz. Explain why this is happening, what the limitation is, and what cable they need to get the full 144 Hz at 4K.

Answer: HDMI 1.4 has a maximum bandwidth of 10.2 Gbps, which is only sufficient for 4K at 30 Hz. Running 4K at 144 Hz requires approximately 48 Gbps of bandwidth. To achieve 4K 144 Hz, the student needs either: HDMI 2.1, which supports up to 48 Gbps and handles 4K 144 Hz natively, or DisplayPort 1.4, which supports 32.4 Gbps and can run 4K up to 120 Hz with Display Stream Compression (DSC). The monitor and GPU must also both support the required cable standard. The old HDMI 1.4 cable is the bottleneck -- replacing it with HDMI 2.1 or DisplayPort 1.4/2.0 will resolve the issue.